



Sending firewall logs to the log relay over TLS

Pointing a FortiGate, Palo Alto NGFW or Meraki MX at the log relay over TLS

Log Relay Manager 1.1.74

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How to point a FortiGate, a Palo Alto NGFW and a Cisco Meraki MX at the log relay with the traffic encrypted. The relay's TLS listener is **tcp/6514** (RFC 5425), the standard port for syslog over TLS; all three vendors use it.

This covers the device side and the one relay setting each needs. It does not cover the relay's own installation, the tenant map or the relay-to-sensor hop — those are in the handbook.

Each device also has a plain-text route, which is worth knowing while you are proving the path works: **udp/514**, **tcp/514** and **tcp/601** (octet-framed) are open on the relay whether or not TLS is configured.

Before you start: three decisions that affect all three vendors

1. The relay's certificate: from the relay's own authority, from yours, or self-signed

The certificate a firewall checks on 6514 is syslog-ng's, and it can come from three places:

Where it comes from	FortiGate	Palo Alto NGFW	Meraki MX
The relay's own certificate authority (recommended)	Yes	Yes	Yes
Your organization's certificate authority	Yes	Yes	Yes
The self-signed certificate the relay made at setup	Yes — import the certificate itself as a CA certificate	No — PAN-OS needs a certificate whose issuer differs from its subject	No in practice — Meraki validates against a CA bundle you upload

The relay's own authority is the simplest route for all three. The console creates it on the relay (Settings → **Certificate authority**), issues syslog-ng's certificate from it, and offers one file, **Download the CA for the firewalls**, that every device imports once. When the certificate is renewed, it is issued again from the same authority and no device needs anything new. The authority's private key is made on the relay, readable only by root, and never downloaded.

If your organization already runs a certificate authority, use that instead and switch the relay's one off (**We use our own authority instead: switch this one off**, at the top of the same tab). Either way, **use one CA for everything** — it makes all three vendors identical.

2. The certificate must name the address the devices send to

Every one of these devices checks the relay's certificate. Palo Alto's own instructions say to set the relay certificate's common name to the address you put in the syslog profile, and Meraki validates the name as well.

So decide now whether devices will send to a **host name** or an **IP address**, and have the certificate carry exactly that — as the common name **and** as a subject alternative name (**DNS:** for a name, **IP:** for an address). A certificate that is right in every other way but does not carry the address a device dials will fail, and the device will say nothing useful.

The relay's authority prefills these for you — the console's own name if one is set, the address your browser is using, the host's name and its addresses — and the issue form lets you add any other name or address the firewalls send to. Check the list before you press Issue.

3. Client certificates are all-or-nothing

The relay has one TLS listener, so whether it demands a certificate *from the device* is a single setting that applies to every device on 6514. The three vendors differ:

- **Palo Alto always presents one.** PAN-OS syslog over TLS is built around mutual TLS.
- **Meraki cannot present one.** Cisco's documentation states client certificates are not supported.
- **FortiGate can, and does not have to.**

If your estate mixes Palo Alto with Meraki, **leave client certificates off** on the relay. The Palo Alto will still connect — it offers its certificate only if the relay asks. Turn the requirement on only when every device sending over 6514 can satisfy it.

Set up the relay

Check the TLS listener is on. It exists only while the relay has a certificate configured:

```
sudo ss -lntp | grep 6514
```

Nothing listening? Turn it on — this is a reload, with no gap in the log flow:

```
sudo stellar-syslog-relay setup --inbound-tls
```

Give syslog-ng a CA-issued certificate (recommended, and required if you have Palo Alto or Meraki).

Either way it is applied by a reload of syslog-ng: no listener closes and nothing in flight is lost, and if syslog-ng refuses the certificate the previous one is put back.

From the relay's own authority — Settings → **Certificate authority**:

1. **Create the authority**, once. Its name is shown as the issuer on everything it signs.
2. Under **2 · Devices sending logs over TLS**, press **Download the CA for the firewalls**. This is the file each device imports below — not the relay's own certificate.
3. Press **Issue the syslog-ng certificate...**, check that the names include the address the firewalls send to, and issue it.

From your own authority — Log relay services → **Security** → **syslog-ng's TLS certificate: Install a certificate and its key** (the certificate, its key and, recommended, your CA's certificate, so the chain is checked before anything changes), or **Or: create a signing request here**, so the private key is made on the relay and never leaves it. Or on the host:

```
sudo stellar-syslog-relay setup --inbound-cert /path/relay-fullchain.pem --inbound-key /path/relay.key
```

If you are using the relay's self-signed certificate instead — FortiGate only — the same card has **Download the certificate**; that file is what the FortiGate imports.

Open the port. tcp/6514 inbound, in the host firewall and in any cloud security group.

Only if every device can present a client certificate, require one: Log relay services → **Security** → **Which devices may connect over TLS**, add the CA that issued the firewalls' own certificates. Optionally, each log source can then name the one certificate its device presents. Or on the host:

```
sudo stellar-syslog-relay setup --inbound-client-ca /path/device-ca.pem
```

FortiGate

Tested on a FortiGate-60F running FortiOS 7.6.6.

The TLS settings are **CLI only** — the GUI's Log Settings page has no port or encryption field, and it will not show you what is really configured.

1. Make the Certificates menu visible

On a fresh unit there is no **System** → **Certificates** entry: it is hidden behind a feature toggle.

System → Feature Visibility → Certificates → Apply.

Or from the CLI:

```
config system global
    set gui-certificates enable
end
```

This only affects the GUI. If you would rather not touch it, everything below can be done from the CLI instead.

2. Check whether VDOMs are enabled

```
get system status | grep -i "Virtual domain"
```

If the System menu shows **Global Resources**, they are. It changes where the settings go:

- **VDOMs disabled** — configure as shown below.
- **VDOMs enabled** — the syslog settings live in the global context (`config global` first) and apply to every VDOM. A single VDOM can override them with `config log setting` → `set syslog-override enable`, then `config log syslogd override-setting`, which takes the same options. **Import the certificate in the same context you configure the syslog settings in.**

3. Import the CA certificate

System → Certificates → Create/Import → CA Certificate → File, and upload:

- the CA certificate, if the relay has a CA-issued certificate — with the relay's own authority, the file from **Download the CA for the firewalls**; or
- **the relay's own certificate file** (`relay.crt`), if it is self-signed. This works because the relay generates it with `openssl req -x509`, which marks it `CA:TRUE` — a FortiGate requirement — and it is its own issuer.

From the CLI instead:

```
execute vpn certificate ca import tftp <filename> <tftp-server-address>
```

There is no setting that ties a CA to the syslog destination. The FortiGate validates the relay's certificate against its whole CA store, and starts logging by itself as soon as a CA that matches is present. If it is missing or does not match, the handshake fails with `Alert (Level: Fatal, Description: Unknown CA)` and **the FortiGate sends nothing, silently** — that is the failure to expect, and it looks identical to a relay that is not listening.

4. Configure the syslog destination

```
config log syslogd setting
    set status enable
    set server "<relay address>"
    set mode reliable
    set port 6514
    set enc-algorithm high
end
```

Order matters: `enc-algorithm` does not exist until `mode reliable` is set. The default port is 514 — set 6514 yourself. Optionally add `set ssl-min-protocol-version TLSv1-2`; leave it out if your build rejects it.

Leave `format` at `default`. FortiGate's native key=value log is what the sensor's FortiGate parser expects; `rfc5424` or `cef` would change what arrives.

If the firewall already logs somewhere else, use `config log syslogd2 setting` (and 3, 4) — the same options.

5. Check what is really configured

```
get log syslogd setting
```

The GUI will not show you the port, and a stale value left over from an earlier experiment is a common cause of "configured correctly, nothing arrives".

Client certificate — normally leave it alone

`set certificate` names the FortiGate's **own** certificate, presented to the relay. You only need it if you turned on the relay's client-certificate requirement; then import a local certificate (**System** → **Certificates** → **Create/Import** → **Certificate**) and name it here.

Palo Alto NGFW (PAN-OS)

PAN-OS is the most demanding of the three, because **syslog over TLS is mutual TLS**: the firewall presents its own certificate as well as checking the relay's. Both must be issued by the same CA. A self-signed relay certificate will not do — PAN-OS requires that a certificate's subject and issuer differ. For SSL syslog, PAN-OS supports **TLS 1.2 only**.

Plan on one CA issuing two certificates: one for the relay, one for each firewall.

With the relay's own authority, it covers the relay's side: import its certificate in step 1. It issues only the relay's certificates, not the firewall's; the firewall's own certificate comes from your organization's CA, and the relay checks it only if client certificates are turned on (Before you start, 3).

1. Import the CA

Device → **Certificate Management** → **Certificates** → **Import**, import the CA certificate (no private key) – with the relay's own authority, the file from **Download the CA for the firewalls** – then open it and tick **Trusted Root CA**.

2. Give the firewall its own certificate

Device → **Certificate Management** → **Certificates** → **Generate** (or import one issued by your CA).

- The **common name must be the IP address of the firewall interface the syslog traffic leaves from**.

This is the step people miss.

- The certificate's private key must be on the firewall – it cannot live on an HSM.
- Open the certificate and tick **Certificate for Secure Syslog**. Without this flag it is not offered in the handshake.

3. Create the syslog server profile

Device → **Server Profiles** → **Syslog** → **Add**

- **Name** – anything.
- **Syslog Server** – the relay's address, matching the name in the relay's certificate.
- **Transport: SSL**. Port 6514 fills in automatically.
- **Format: BSD, Facility** as you like.

4. Attach the profile – nothing is sent until you do

This is the single most common reason a correct PAN-OS setup sends nothing:

- **Traffic, Threat, URL and the other session logs: Objects** → **Log Forwarding**, create or edit a profile, add a match list pointing at the syslog profile, then apply that log-forwarding profile to your **security rules**.
- **System, Config, HIP and User-ID logs: Device** → **Log Settings**.

Then **Commit**.

5. Set the firewall's time zone to UTC

Device → **Setup** → **Management** → **General Settings** → **Time Zone**.

PAN-OS syslog has nowhere to state a UTC offset – not in its header and not in its CSV timestamps – so a firewall on local time has its events read as UTC and they land in the platform hours in the past, where a search over the last 15 minutes will never find them. The other two vendors each state what they mean, which is why this only bites Palo Alto.

Revocation checking

PAN-OS validates the certificate chain with OCSP or CRL where the certificates carry those extensions, and **a revocation check that fails cannot be bypassed**. If your CA issues certificates carrying a CRL distribution point or an OCSP responder, the firewall must be able to reach it.

Cisco Meraki MX

Meraki MX supported **UDP only** until recently. **MX26.1 and later support TCP and TLS**, so check the firmware first — on an older release the TLS options simply are not offered and UDP is your only route.

Two Meraki-specific limits to plan around:

- **No client certificates.** So do not turn on the relay's client-certificate requirement if a Meraki is sending to it.
- **Switch roles cannot use an encrypted entry.** Configure only appliance and/or wireless roles on a TLS syslog server entry; switch logs need a separate, unencrypted entry.
- Z-series Teleworker Gateways do not support encrypted syslog.

1. Upload the CA

Organization → **Certificates**, upload your CA bundle — with the relay's own authority, the file from **Download the CA for the firewalls**.

This step is not optional for a private CA: with no bundle selected, Meraki validates the relay's certificate against the Common CA Database — the public trust store — and a privately issued certificate will be rejected.

2. Add the syslog server

Network-wide → **Configure** → **General** → **Syslog servers** → **Add a syslog server**. (On a single-device-type network this section is called **Logging** rather than **Reporting**.)

- **Server IP or host name** — the relay, matching the name in its certificate. Host names are supported from MX26.1.
- **Port** — 6514.
- **Transport / encryption** — TCP with TLS, and select the CA bundle you uploaded.
- **Roles** — Appliance event log, Appliance flows, Appliance URLs, and IDS security alerts as required. Flows is the high-volume one; turn it on deliberately.

Save. Meraki applies it to the network without a reboot.

3. If the firmware is older than MX26.1

Send over **udp/514** instead, and keep that leg inside the tenant network or a site-to-site VPN. Meraki's timestamps are epoch seconds, so there is no time-zone trap to worry about on this one.

Proving it works — any of the three

Work outwards from the device.

1. Is anything arriving at the relay at all?

```
sudo tcpdump -ni any port 6514
```

Or use **Packet capture** in the console, which can be narrowed to one log source and will follow that device's logs out to the sensor as well.

2. Did the handshake succeed, or is the device being refused?

```
sudo journalctl -u syslog-ng -n 50 --no-pager
```

A certificate the relay rejects, or a device that gave up on the relay's certificate, shows here. On the device side: FortiGate `diagnose sniffer packet any 'host <relay> and port 6514' 4`; Palo Alto has the system log; Meraki reports nothing, so judge it from the relay.

3. Is the relay recognising the device?

Console → **Log sources & tenants**. The device's "last seen" dot goes green within a minute of its first log. This needs the copy of forwarded records to be on (**Sensors** → **Copy of forwarded records**); if it is off, the console will tell you so rather than claim the device was never seen.

4. Did it reach the platform?

Search for the device in Stellar Cyber. Give it two minutes and widen the time window past the last 15 minutes — records can take a minute or more to become searchable, and generated firewall traffic events land in the Traffic index, not Syslog.

If the device shows as connected but no logs appear

- **Palo Alto**: the server profile is almost certainly not attached to a log-forwarding profile, or that profile is not applied to any security rule.
- **FortiGate**: check `get log syslogd setting` for a stale port, and confirm logging is actually enabled for the traffic you expect.
- **Meraki**: check the roles on the syslog entry — an entry with only event-log roles sends very little.

If nothing arrives at all

Work through, in this order: the port is open end to end (host firewall, cloud security group); the relay is listening on 6514; the CA is imported on the device; the relay's certificate names the address the device is configured with. The last one is the quiet failure — everything looks right and the handshake is refused.



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